

Gunter & Scott 1990 — Inventory of Definitions and Theorems

Source: C. A. Gunter and D. S. Scott, “Semantic Domains,” *Handbook of Theoretical Computer Science* Vol. B, North-Holland, 1990, pp. 633–674 ([./papers/Gunter Scott 1990.pdf](#)).

The work list for the Lean formalization: every definition and every one of the paper’s **30 numbered results** (Theorems / Lemmas / Proposition 1–30), in paper order, matched to its Lean equivalent.

Progress (as of r0037, 2026-0807)

#	Quantity	Done	Remaining	Of
1	Definitions to define	all ≈ 13 — the computable function landed in r0031; see the note below on D^∞	0	≈ 13
2	Numbered results complete	24 (Thm 1, Thm 3, Lem 4, Lem 5, Thm 6, Thm 7, Lem 8, Lem 9, Lem 10, Thm 11, Thm 12, Lem 13, Thm 14 , Prop 15, Lem 17, Lem 19, Lem 20, Thm 21, Thm 22, Lem 23, Lem 24, Thm 25, Thm 26, Thm 27) — the two in bold landed in r0036, and Thm 27 is now <i>unconditional</i> ; three of the 24 carry a qualification, see row 2c	4 (Thm 18, Lem 28, Thm 29, Lem 30 — Thm 18’s steps 2 and 3 are proved, Lem 28 is 2 of 9 at the paper’s own notion, Thm 29’s first sentence is proved and V is built with $V \equiv V^+$; see row 2d)	29
2a	— resolved by refutation	Thm 16 — fully characterized as of r0034. The algebraic-lattice conjunct is proved (r0028); the $Fp(D) \hookrightarrow (D \rightarrow D)$ embedding conjunct is false , kernel-checked (r0032); and the conjunct does hold under a named hypothesis, proved as Section62.thm16_positive with thm16_positive_isEmbeddingProjectionPair (r0034). The result is settled in all three directions, but it is still not a proof of the paper’s sentence, so it is counted in neither column. The row-2 arithmetic is therefore $24 + 4 + 1 = 29$	—	—
2c	— qualifications on three of the 24	Lem 9 : four conjuncts hold as printed; items 3 and 5 are false as printed and are kernel-checked negations (lem9_3_printed_false, lem9_5_printed_false), with the corrected laws proved. Lem 17 : 10 of 10, but the $\rightarrow$ and $\circ\rightarrow$ conjuncts carry [BoundedComplete β] from the step-function decomposition — stronger than the paper states, and §6 exists precisely to avoid bounded completeness. Thm 26 : proved with $hs : \forall i, 0 < s i$; the theorem is false for a signature admitting arity 0, which the paper explicitly allows — that argument is prose in the docstring, <i>not</i> kernel-checked. Lem 10 carries no qualification	—	—

#	Quantity	Done	Remaining	Of
2d	— status of the four remaining, after r0042	Superseded on the first clause: the development has had sorry 0 since r0042 and Theorem 18 is proved (row 6). Retained for the Corollary 1.36 history below. As written: Thm 18 was then the development’s only remaining sorry, and rested on one named proposition. Jung’s Corollary 1.36 is proved (r0042, JungCor136) — not by his route, which goes through Proposition 1.22 and a retraction pair, but by indexing the approximating family below <code>cap e</code> rather than below <code>id</code>, so the deflation condition is built into the index and the object r0037 measured as unavailable is never wanted. What remains is <code>JungNets.Thm137</code>, and r0042 reduced it further: <code>Iwamura.thm137Chains_iff_thm137</code> and <code>Thm18.thm18_of_thm137Chains_and_cor136</code> show only chain infima are ever spent, and <code>JungBicomplete.hasChainInfima_of_forall_isOrdinalCodirected</code> shows Jung’s Corollary 1.3 is <i>free</i> for chains. Thm137Chains now factors as the retraction onto $D' = A \cup \alpha^{\circ p}$ plus the g_β family, and nothing else. The assembly is a library theorem (<code>ScottDomains.Thm18</code>), not only a script, and <code>thm18</code>’s statement was audited against printed page 33 and found correct in all four constituents. Lem 28 is 7 of 9 over the paper’s own Dyadic.U with no hypothesis ($\rightarrow$, $+$, $\times$, $\otimes$, $+$, $\otimes$, $()^\perp$), up from 0 before r0037; $()^\#$ and $()^\flat$ remain. Thm 29’s second sentence is reduced to the single proposition <code>LemThirty.Thm29Normal</code>, kernel-checked in ~60 lines. Lem 30 is stated at ten conjuncts over V and remains 0 of 10, but the §7.3 schemes are measured to be generic in the carrier, so it is ten instantiations plus ten retraction pairs, not ten fresh proofs	—	—
2e	— unspecified: paper properties with no Lean statement at all	was 62 of 239 when r0040 measured it (five agents over all five sections; see ../analyses/property-coverage.2026-0808-11:59.orchestrator.md). r0041 closed about 40 of them by building the four constructions they clustered on: the flat <code>cpo</code> (<code>Flat</code>, an inductive — <code>WithBot</code> keeps the carrier’s order and <code>WithBot Bool</code> is <i>linear</i>, where the flat order needs the two non-$\perp$ points incomparable), the morphism-level algebra at 11 of 11 (<code>Morphism</code>), the powerdomain functor $f_\sharp/f_\#/\flat$ with both functor laws (<code>PowerdomainMap</code>), and instantiated effective presentations (<code>Effective/</code>)	26, and zero of them numbered. This is measured, not derived — r0043 re-checked all 62 rows against the tree (five agents, own-area N rows only; see ../analyses/property-coverage-remeasure.2026-0808-16:55.orchestrator.md). 36 closed, 26 remain; the “≈ 22” that stood here was $62 - 40$ by subtraction and was low by 4. The 36 break down 33 $S+P$, 2 $S\neq$, 1 $S+H$. The residue is concentrated rather than spread: 13 of the 26 are §7 alone, where re-measurement moved <i>nothing</i> — r0041 made $X \cong N\perp + (X \rightarrow X)$ writable for the first time by supplying <code>Flat.NatBot</code>, and no statement writes it, so that round changed expressibility, not statedness. The other 13 are three missing carriers ($\mathbb{R}$, $\mathbb{Q}$) with two meta-claims blocked behind them, $N\perp$’s two function spaces, three order facts, a Figure 3c poset nobody has built, and one “subset” claim without $\subseteq$	239

#	Quantity	Done	Remaining	Of
2f	— stated in prose only: claims the devel- opment asserts but never puts under the kernel	3 — (1) Theorem 26 is false for a signature admitting arity 0; (2) Colimit.Thm29Second is false without [Domain E], where countable_compacts_of_reflects is checked but the step from it is not; (3) the $\otimes/\otimes$ counterexample refuting Lemma 28’s closure reading, 30 lines at CombinatorRep.lean:504–533 whose own text says “a hand computation, not Lean-checked”	all 3 are cheap to put under the kernel; the lem10_smash and Theorem 16 precedents say to do it	—
2g	— why these two rows exist	sorry counts holes in <i>stated</i> theorems. A result nobody wrote down produces no warning, no count and no build signal, so rows 6 and 2e measure different things and neither substitutes for the other. A development can reach 0 sorry while leaving paper results unstated , and this one is two results away from exactly that	—	—
2h	— mis-specified: stated, but not as the paper states it	13 defects of 18 S≠ rows (r0044 agents 1–2; see ../analyses/specification-defects.2026-0808-21:05.orchestrator.md). Split 12 under-specified, 1 incorrectly specified, 5 deliberately divergent — the last being repairs of the paper’s own printed defects, which are correct work and not counted. The S≠ label had merged all three, so the 18 previously quoted overstated our defects by 39%. The structural finding: 9 of the 12 are the added-binder shape , stating the paper’s conclusion under <i>stronger hypotheses than the paper assumes</i> . Only one is a missing conjunct. r0040 and r0043 both hunted missing conjuncts, which is why this went unseen — an added binder reads as faithful transcription until you compare them	13	18

#	Quantity	Done	Remaining	Of
2i	— vacuously specified: labelled S+P, hypotheses unused	7 — 6 from r0044 agents 3–5, plus one from r0045 by a second and independent mechanism: Effective.PreservesRecursivePresentation quantifies over a γ unrelated to its α and β, so the identity closes it in one line, and its closure over γ is false. That vacuity lives in the quantifier structure, not a field type, so a reviewer checking only for Classical.dec passes it; both mechanisms are now a two-question check in StructuresVsTypeClassesVsPropsInLean4.md. r0047 swept for the second mechanism and found zero further instances — scripts/a3-r47-qvac-body.lean builds the parameter graph and flags a conclusion parameter whose connected component contains no hypothesis parameter, returning 1 hit in 2,111 declarations, the known one, at 100% precision with recall cost 0 against a looser criterion scoring 6.7% (all 15 of its hits adjudicated by reading). Five controls print on every run, one of them a cross-instrument control showing r0045's a5-freehyp reports 50 free Prop rows and says nothing about this mechanism — confirming <i>by measurement</i> that the field-based sweeps would have passed it. The known instance was itself restated in r0047 to quantify over an operator, which is what the paper's sentence quantifies over; that restatement is a strengthening, running opposite to r0046's, and preservesRecursivePresentation_arrowOp_iff proves the new claim at the arrow equivalent to Theorem7ArrowRecursive — a relationship the old docstring asserted but the free-carrier statement could not even express. The first mechanism: Classical.dec inhabits EffectivePresentation's decidability fields at no cost, so <i>every</i> domain has one and any theorem quantifying over it is vacuous — see StructuresVsTypeClassesVsPropsInLean4.md. #lint unusedArguments alone finds zero of them: it exempts <code>_</code>-prefixed binders, and both originally-known cases are named <code>_d/_e</code>. Three complementary instruments were needed — unused-in-proof-term, used-but-free (which found all 4 new ones), and deletion-plus-reproof. A separate 26 declarations are over-hypothesized, proving more than they claim; that is the opposite sign and is deliberately excluded	6	1,773

#	Quantity	Done	Remaining	Of
2j	— undis- charged: claims stated as Prop- valued defs that no theorem proves	7 open after r0047 (8 strict) — unchanged in number by that round, and that is the honest headline (see ../analyses/close-the-seven.2026-0809-12:35.orchestrator.md). Three of the seven now mean something different. Thm29Normal's route is a dead end: Gunter's Lemma 24 at $M(A)$ is <i>proved</i> (R47.Agent1.lemma24_MPair, the first proof anywhere — the printed Lemma 24 is not about $M(A)$, and the $M(A)$ form is a remark with no theorem and no proof), but <code>HasNormalRealizations Ainf</code> is refuted, so r0046's sound implication has an unsatisfiable hypothesis. StepFunctionsDecidable's subject has a kernel-checked defect: <code>IsCompactElement (ofPairs Q)</code> is <i>not</i> the boundedness test, since <code>sSup</code> on <code>ScottHom</code> is total and reads a junk value on an inconsistent Q — our transcription error, not the paper's. The domain theory it needed is proved (<code>bddAbove_steps0f_iff</code>), and restating over the consistency-guarded enumeration — machinery already built — closes four claims, not three. Separately r0047 removed <code>[BoundedComplete β]</code> from six declarations including Domain (ScottHom α β), which Goal A cannot see at all. Also the S+H unfinished-proof count was 16 and is 12 — four rows had closed in r0042–r0045 and the tally was never decremented. The prior figure was 7 after r0046, itself down from 10, re-derived on the merged tree (see ../analyses/zero-props-zero-false-prose.2026-0809-11:40.orchestrator.md). Three are now refuted and therefore resolved — <code>Colimit.Thm29Second</code>, <code>PRep.Lemma28</code>, <code>LemThirty.Lemma30</code>, the last having been argued false in prose across two rounds and never proved so until <code>not_forall_lemma30</code>. The metric itself was defective and is fixed: a refutation produces $\neg \text{Foo}$, not Foo, so a claim proved false still scored as undischarged; <code>a6-summarize.py</code> now reports refuted as a third status and flags resolutions holding only at a parameter instance — which is why the strict count is 8, <code>Effective.PreservesRecursivePresentation</code> being resolved only at $\gamma := \alpha$ and $\gamma := \text{Flat } \mathbb{N}$. Thm29Normal was never blocked on the unobtainable [Gun87]: the proof is in <code>papers/Gunter 1987 Universal Profinite Domains.pdf §5</code>, already cited at <code>BifiniteUniversal.lean:47</code> for the construction, and r0046 reduced the claim to one named input, Gunter's Lemma 24 at $M(A)$. <code>StepFunctionsDecidable</code> is blocked on one domain-theoretic lemma — deciding <code>IsCompactElement (ofPairs Q)</code> — not on the two recursion-theoretic obstructions previously recorded, both of which were measured real and non-blocking. The prior figure was 10 after r0045, itself down from 19, re-derived by re-running the detector over the built environment rather than by subtraction (see ../analyses/discharge-	19	88

#	Quantity	Done	Remaining	Of
3	Unnumbered prose claims	146 stated by the paper (r0040, measured section by section: §2/§3 45, §4 37, §5 17, §6 19, §7 28 — these sum to 146), of which the development proves 70 and states in some form 86. The figure here was 91 until r0044, and 91 was the <i>numbered-results</i> count (“91 of 93 numbered conjuncts are stated”) transcribed into the prose row; it exceeds the maximum possible by 5, since only 86 prose claims are stated at all. The row’s arithmetic is also structurally unsound and always was: $146 - \text{proved} = \text{unstated}$ assumes the categories are complementary, but S+H, S≠ and P account for 16 prose rows sitting between “proves” and “unstated”, so no assignment of three numbers can close it. Separately, the figure before that was 13, and it was not a count of the paper — the list was assembled from claims the development proves, every entry naming a Lean declaration, so a claim with no declaration could never enter it, and it was then used as a coverage denominator. r0038’s correction of that list (strike one claim the paper does not make, add two §4.1 claims it does) stands and is folded in	60 unstated. This 60 is correct and only looks inconsistent because of the cross-reference: it is the prose-only N sum (62 less the two <i>numbered</i> N rows), whereas row 2e’s 62 ranges over all 239 properties	146
4	Mathlib foundations reused	12	—	12

#	Quantity	Done	Remaining	Of
5	Modules / lines / theorems	117 / 43481 / 2019 — measured on main after r0047, Audit/ included; build 1364 jobs, 0 errors, 0 warnings. The environment reports 2,114 theorems and 4,089 constants, which measure different sets and must not be reconciled with this row. Previously 109 / 40830 / 1911 after r0046; build 1356 jobs. r0045 added five modules (the discharge round) and r0046 five more, among them <code>A2Thm29Universal.lean</code>, which reduces Theorem 29’s normality claim to a single named input, and <code>Effective/A3StrictRecursive.lean</code>, which supplies the enumeration of $K(D \rightarrow E)$. Three other counts measure different sets and must never be reconciled with this one: the environment reports 2,009 theorems (adding Prop-valued class instances) and 3,908 constants (adding defs, structures, projections and equation lemmas). Previously 99 / 36275 / 1735 with r0041 and r0042 merged. r0041 added 16 modules and r0042 six, among them <code>Iwamura.lean</code>, which carries Iwamura’s lemma and Markowsky’s theorem — neither has ever been in Mathlib. Previously 72 / 27892 / 1298 with r0037 merged, re-measured after r0038 by <code>scripts/lean-decls.py</code>, which <code>counts.sh</code> now calls: the <code>grep</code> rule it replaced counted declarations inside <code>/- ... -/</code> block comments and <code>docstring</code> prose lines beginning “theorem”/“lemma”, and missed protected theorem, so the previously reported 1308 was an over-count by 10. The corrected figure is within one of r0038’s entirely independent per-declaration enumeration, which totalled 1297	—	—
5a	— plus the r0038 audit modules	<code>ScottDomains/Audit/</code> adds 6 / 670 / 28 — the kernel-checked equivalences proving the duplicate pairs the audit found. Package totals with them: 78 / 28562 / 1326. They are audit artifacts, not development, so row 5 excludes them	—	—

#	Quantity	Done	Remaining	Of
6	sorry in the development	0 — and still 0 after r0046 , alongside 0 axiom declarations and 0 constants naming sorryAx , each confirmed by two independent instruments. Read this row with 2j: sorry counts holes in proofs, and cannot see a claim nobody attempted — 7 such claims remain, recorded as Prop-valued defs by a deliberate convention that is better than a sorry (it asserts nothing and propagates no sorryAx) but was never recorded where a count would find it. As first established: the development is sorry-free as of r0042 . thm18 is proved: PropertyM.thm18_of_cor136 applied to JungCor136.fixedPointOfCompactDeflationIsComplete, and #print axioms ScottDomains.thm18 gives [propxet, Classical.choice, Quot.sound] with no sorryAx. Jung’s Theorem 1.37 was not needed — PropertyM.forall_hasCompleteMub derives property m from IsAlgebraic (ScottHom D D) and countability of $K(D \rightarrow D)$, hypotheses Theorem 18 already carried, following Spreen 2005 Lemma 5.8. Five rounds treated Theorem 1.37 as the blocker; it was a strictly stronger statement than the proof needed	Formerly 1: thm18 at Skeleton/Section6.lean:202. r0034 took this from 8 to 2 and r0036 from 2 to 1 by proving Theorem 14. r0042 established that closing it is a two-line edit : the planned one-line exact was an <i>import cycle</i> — Skeleton.Section6 sat inside JungFinite’s own import cone because FinitaryProjectionPoset imported the skeleton solely for IsClosure — and moving eight closure declarations to ScottDomains/Closure.lean removed it, which also restores the Skeleton/statements-only convention. Every other statement in the development is kernel-checked, and nothing in the development depends on thm18 — the only mention outside its own file and Section62.lean’s obstruction write-up is a docstring reference, so no completed result routes through sorryAx	—

Round r0042 — the development is sorry-free. Five agents; thm18 closed, taking the count $1 \rightarrow 0$ and the numbered results to **25 of 29**. The route that closed it is not the one five rounds had assumed: **Jung’s Theorem 1.37 was never needed**. Spreen 2005’s Lemma 5.8 proves property m *at a pair* directly, and its σ turned out to be JungSFP.jungFun with a level-indexed top region — machinery the development already had. The load-bearing measurement was that two existing theorems asked for property m at every finite set while using it only at $\emptyset$ and at pairs; restating them against what they use lifted the pair case to the whole. Also **Jung’s Corollary 1.36 proved; Iwamura’s lemma and Markowsky’s theorem formalized**, neither ever in Mathlib; the Theorem 18 assembly made a library theorem instead of a script; and thm18’s statement audited against the paper and found correct in all four constituents — the first time that had been done, after five rounds of trying to prove it. Five corrections to Jung’s own write-up were recorded, the sharpest being that **his retraction r is a closure, not a projection** (his formula gives it; he never says so, and without $\text{id} \leq i \circ r$ the proof establishes nothing about D), and that **r is not total** — its infimum is over an empty set for points not below the chain’s largest element, so his two-case formula needs a third. Net: sorry still 1, and what remains of Theorem 18 is one construction rather than a research programme.

Round r0041 — stating what was never written down. Five agents against the 62 properties r0040 found unstated, split by *missing construction* rather than by paper section, because the gaps clustered on four absent objects. All four landed. Two numbered results were repaired rather than extended: **Theorem 7’s second and third sentences** (closing the last numbered gap), and **Theorem 3**, which had been proving *uniqueness of an operator never shown to exist* — UniformFixedPoint.lean declined to assert existence, and Kleene.theorem3_existsUnique now supplies it. Two more printed defects found: §5.2’s clause 1 of the convex pre-order is **false as printed**, and §6’s “no least upper bound for $\{u, u'\}$ ” is a typo for $\{u, v\}$.

Rounds r0038 and r0039 changed no paper coverage, and are recorded here only because they corrected numbers this file reports. r0038 was an audit — six agents classifying every theorem against the paper, deleting nothing; its consolidated result is [../analyses/theorem-audit.2026-0808-10:04.orchestrator.md](#), and the headline is that speculative API sits at $\approx 3.5\%$ of 1297 live declarations against r0020’s 3%, so the $\sim 13 : 1$ theorems-per-property ratio is the cost of formalizing a paper that elides its own foundations rather than accumulated bloat. What it did find is **31 declarations in ~ 21 duplicate pairs spanning module boundaries**, none of them visible to lake build because nothing imports both sides. It also corrected rows 3 and 5 above. r0039 is redrawing the paper’s diagrams as TikZ, one PDF each, in [../GunterScott90Images/](#).

Round r0037. Five agents at the last four open numbered results; all five landed and merged. **No numbered result was completed** — the count stays at 24 of 29 and sorry stays at 1. What changed is the size of what remains, and the honest headline is a set of reductions rather than a completion. 72 modules, 27866 lines, 1308 theorems. Build 1223 jobs, 0 errors, 0 diagnostics, 0 warnings beyond sorry. Composition check: all six new modules import into one environment with no clash, every headline theorem on [propxet, Classical.choice, Quot.sound].

#	Stream	Landed
1	JungFinite.lean	Theorem 18's step 4 and its assembly — Jung's Lemma 1.29, König's lemma graded by $\mathbb{N}$ in place of Rado's Selection Theorem (Jung proves Rado over an arbitrary index set by Tychonoff, but Lemma 2.2 applies it only at $I = \mathbb{N}$; Mathlib's <code>nonempty_sections_of_finite_inverse_system</code> needs a functor out of a category plus the product topology, while the elementary proof is ~100 lines with no topology), Lemma 2.2, and <code>thm18_of_propertyM</code>
2	JungNets.lean	Theorem 1.37's consequence, and the correction that it is not what the plan said. Jung p. 50 reads "A <code>dcpo</code> with continuous function space is bicomplete ", not "has property <code>m</code> "; property <code>m</code> is a separate inference inside the proof of Theorem 2.3 which Jung never proves. That inference is now proved in full — <code>exists_minimal_upperBounds_le</code> by Zorn downwards — and <code>JungSFP.lemma217</code> 's hypothesis is discharged by application. Theorem 1.37 itself is the named <code>Prop Thm137</code>
3	PRepFun.lean	Lemma 28's $\rightarrow$, $*$ and $\otimes$, plus the two <code>Domain</code> instances they needed and the development did not have: <code>strictHomDomain</code> (40 lines — the strict function space is downward closed in $D \rightarrow E$, so its compacts are literally those of $D \rightarrow E$ below) and <code>smashDomain</code> (120 lines — no such embedding; needs the compactness criterion plus a cofinality argument discarding approximants with a $\perp$ coordinate)
4	PRepSum.lean	Lemma28AtU , four lines from <code>Atomless.thm27</code> ; $+$ and $\otimes$; and <code>isAlgebraic_coalescedSum/domain_coalescedSum</code> , closing a gap no round had recorded
5	LemThirty.lean	Lemma 30 stated at ten as one conjunction with <code>lemma30_of</code> , and <code>lemma30_iff_lemma28_and_plotkin</code> making "the same nine plus $() \vdash$ " a theorem; <code>plotkinOp</code> ; and Theorem 29's second sentence reduced to the single proposition <code>Thm29Normal</code>
—	Lemma28AtU.lean	the orchestrator's join : agent3's three conjuncts lifted to <code>Dyadic.U</code> by agent4's <code>pairAtU</code> , taking Lemma 28 at U from 4 of 9 to 7 of 9

Two stale cross-stream claims, each true when written and false on merge. `r0036`'s agents ran concurrently and none could see another's result, so `PRep.lean` recorded `Lemma28AtU` as "blocked one level below this file" while agent3 was removing that block, and `Colimit.lean` recorded the nine operators as "not present as functions $Cpo \rightarrow Cpo$ at all" while agent4 was defining all nine. Both were found by reading the merged tree, not from any report. **A cross-stream reconciliation pass at merge is now a standing orchestrator step**, and `r0037`'s own join module is the first artifact of it.

Two library gaps closed, neither previously recorded. `Smash`, `CoalescedSum` and `SeparatedSum` were never proved algebraic anywhere — they carried Lemma 10 and Lemma 17 and nothing else. Found by agent5, relayed mid-round, and closed by agent3 and agent4 rather than carried as hypotheses.

Three corrections to the r0037 plan, continuing the pattern (`r0034`: four, `r0036`: three): Theorem 1.37's statement, Lemma 1.29's "empty set *and* each pair" (the clause is not redundant — an infinite antichain refutes it without, though over $K(D)$ in a `cpo` it is free), and that step 4 does **not** need `Domain.countable_compacts`, the $\mathbb{N}$ -grading coming from the `U`-iteration instead. Countability is still indispensable to Theorem 18 and is still spent exactly once, in `JungSFP.lemma217`. Two streams reported **no** correction — agent3 and agent4 each re-read §7.3 and found the operator list right.

Round r0036. Five agents at the six then-open numbered results; all five landed and merged. sorry $2 \rightarrow 1$; numbered results $22 \rightarrow 24$ of 29; 66 modules, 23596 lines, 1119 theorems. Build 1217 jobs, 0 errors, 0 diagnostics, 0 warnings beyond sorry. Composition check: all five new modules import into one environment with no name clash, and every headline theorem depends only on [`propxt`, `Classical.choice`, `Quot.sound`].

#	Stream	Landed
1	SFP.lean	Theorem 14 in both directions — the last sorry but one. Of <code>r0034</code> 's four measured gaps, two were real (the bridge <code>im(p_N) = N</code> for finite normal <code>N</code> , and two finite-directed-set lemmas) and two were false constraints : the <code>Fp(D)</code> machinery is never entered, because on a finite image <code>im(p) ∩ K(D) = im(p)</code> ; and leastness of <code>id</code> is proved directly in <code>ScottHom α α</code> , not transported from <code>⊥ (Fp α)</code> . One ingredient no earlier note had listed: <code>M</code> must be nonempty , since <code>IsCompactElement</code> quantifies over nonempty directed sets while Mathlib's <code>DirectedOn</code> is vacuous on $\emptyset$

#	Stream	Landed
2	JungSFP.lean	Theorem 18's steps 2 and 3 — lemma213, thm214 (the bifurcation) and lemma217 (the uncountable family). What unblocked three failed rounds was not a tactic but a missing lemma: the development states minimal upper bounds <i>relative to</i> $K(D)$, while every one of Jung's minimality steps applies them to a bound not known to be compact. <code>isCompactElement_of_minimal_upperBounds</code> (Jung's Prop 1.9 for <code>IsAlgebraic</code>) bridges the two, and without it <code>f_A</code> 's monotonicity is false. IsLDomain proved unnecessary — Lemma 2.17 uses only condition (vii) of Jung's Theorem 2.10, so <code>HasTwoMubBelow/HasAtMostOneMubBelow</code> are defined directly and the six equivalences the route never passes through are not formalized
3	Atomless.lean	Theorem 27, unconditionally. <code>IsNormallyRepresented</code> $\mathfrak{r}(\text{compacts } D)$ is proved, so <code>Atomless.thm27</code> carries no hypothesis. Neither Vaught's theorem nor a Boolean algebra is needed — two claims in <code>Dyadic.lean</code> 's docstring were false and are corrected. What Theorem 27 consumes is three properties of a map $\psi : K(D) \rightarrow U_0$: order embedding into the superset order, $\psi \perp = [\emptyset, 1]$, and $\psi a \cap \psi b$ empty when $\{a, b\}$ is unbounded and $\psi(a \sqcup b)$ when bounded. The third is also what makes <code>range</code> ψ normal, so the paper's unproved final clause is four lines. <code>Legal</code> demands that a join drag in no earlier <code>enum</code> i the branch does not carry, which is what keeps ψ a finite union of intervals
4	PRep.lean	Lemma 28 at the paper's own notion — $\times$ and $() \perp$ p -representable over $Fp(U)$, plus <code>isFinitaryProjection_sSup</code> , the keystone giving pointwise least upper bounds in $Fp(D)$. The three r0034 conjuncts are kernel-checked not to transfer: <code>Combinator.Rettracts</code> gives $\text{id} \sqsubseteq \text{gr} \circ \text{fn}$ where the projection scheme needs $\text{gr} \circ \text{fn} \sqsubseteq \text{id}$, and <code>gr_fn_eq_of_both</code> proves holding both forces $U \cong V$. Position 3 of 9 at a wrong notion $\rightarrow$ 2 of 9 at the right one. $\otimes$ and $\oplus$ are no longer refuted: a projection has $p \perp = \perp$, so r0034's three-chain counterexample does not apply
5	Colimit.lean	V with Domain V, IsBifinite V and V $\approx_o$ V* — Theorem 29's fixed point, built as the Antisymmetrization of a pre-order on Σn , <code>Stg n</code> with <code>Nat.leRecOn</code> transport and zero casts. Lemma 30 is statable for the first time

Three corrections to the r0036 plan, all from agents reading the PDF. The plan's Lemma 28 operator list was wrong in two places — it inserted $() \mathfrak{r}$, which §7.4 opens by saying *cannot* be representable over U , and dropped $\ast$, the strict function space; `CombinatorRep`'s r0034 docstring already had the correct nine and the plan drifted from the file. Lemma 30 has **ten** conjuncts, not nine — Lemma 28's nine plus $() \mathfrak{r}$, which is the whole point of §7.4. And the plan's route for V , taken from `BifiniteUniversal.lean` and from §7.4's own sentence, says the colimit is along η ; **that colimit is not a fixed point of M** , kernel-checked as `Colimit.stgEmb_ne_mk_eta` — a third printed defect in §7.4, alongside the two r0034 recorded. The correct chain applies M to the previous connecting map; the two first differ at stage $1 \rightarrow 2$, where both values lie among §7.4's five elements of I^{++} , so Figure 4 does not discriminate them. Stage sizes are unaffected, so the 1, 2, 5, **20** count still selects `MPair.le` over the Smyth reading's 21.

Round r0034. Six agents, all landed and merged. sorry $8 \rightarrow 2$; numbered results $18 \rightarrow 22$ of 29; 61 modules, 19497 lines, 906 theorems. Build 1137 jobs, 0 errors, 0 warnings beyond sorry. Composition check: all fourteen new modules import into one environment with no name clash, and every headline theorem depends only on [`propext`, `Classical.choice`, `Quot.sound`].

#	Stream	Landed
1	Isomorphism/ (6 modules)	Lemma 9 — all six laws as named $\approx_o$, plus <code>lem9_3_printed_false</code> and <code>lem9_5_printed_false</code> , kernel-checked negations of the two misprints. <code>Skeleton/Recovered.lean</code> 7 sorry $\rightarrow$ 1
2	ClosureProperties/ (4 modules)	Lemma 10 at 7 of 7 and Lemma 17 at 10 of 10 , each now stated as <i>one</i> theorem — a conjunction over the paper's own operator list — so the conjunct count is kernel-checked rather than prose. Also <code>liftIsAlgebraic/liftDomain</code> : $D \perp$ was only a cpo, so $D + E$ had not been statable
3	Dyadic.lean	§7.3's universal domain U as the ideal completion of the dyadic half-open intervals under <i>superset</i> , with $K(U)$, <code>Domain U</code> by <code>inferInstance</code> from Theorem 11, and <code>BoundedComplete U</code> ; Theorem 27 proved from <code>IsNormallyRepresented</code>

#	Stream	Landed
4	Combinator.lean, CombinatorRep.lean	Theorem 26 — the Engeler-style embedding result, <i>not</i> equation-solving; 3 of Lemma 28’s operators over an abstract Retracts interface; the counterexample refuting the closure reading of $\otimes$ and $\oplus$
5	BifiniteUniversal.lean, PRepresentable.lean	Theorem 29’s first sentence; p-representability over $\text{Fp}(U)$ with <code>eq_id_of_mem_Fp_of_mem_Fc</code> proving it distinct from <code>IsRepresentable</code> over $\text{Fc}(U)$; two kernel-checked defects in §7.4’s printed pre-ordering
6	Section62.lean	thm16_positive and its embedding–projection pair; the Theorem 18 obstruction , written as the deliverable — <code>thm18</code> untouched

Four corrections to the r0034 plan, all found by agents reading the PDF rather than the plan. Theorem 26 is an Engeler-style embedding result, not “combinators solving a signature’s equations”; Lemma 28 lists nine operators at p-representability over $\text{Fp}(U)$, not seven at the closure reading; Theorem 14’s obstacle is mathematical (Plotkin’s SFP characterization), not definitional as the plan asserted; and `prop15/thm22` are not on Theorem 27’s route at all — §7.3 cites no earlier result of the paper.

Round r0032. Five agents; three landed, two were still running when this was written.

#	Stream	Landed
1	Universality.lean	Lemma 24 and Theorem 25 , with <code>thm25_powerset</code> : $\text{P } N$ is universal. Proved at <i>cpo</i> strength — no step spends algebraicity or countability, so it is stronger than the paper’s statement
2	IdealCompletion.lean, Powerdomain/BoundedComplete.lean	the idealSup repair , and bounded completeness reinstated for D_b and $D_\#$
3	FinitaryProjectionEmbedding.lean	Theorem 16’s embedding conjunct refuted
4	ContinuousAlgebra.lean	Theorem 12 for all three powerdomains, existence <i>and</i> uniqueness, 1254 lines, 0 sorry
5	Skeleton/Recovered.lean, docs/StatementRecovery.md	Lemma 9 and Theorem 14 recovered from the PDF — both were recorded as “not statable”

pdftotext drops glyphs, and four inventory rows undercounted because of it. The paper’s 18 embedded Type 3 fonts carry Custom encoding with no ToUnicode map: codes $\geq 0x20$ leak through as ASCII, and **codes below $0x20$ vanish silently**. Those are standard TeX positions, pinned by the leaked ASCII ($0x21 \rightarrow \rightarrow$, $0x3F \rightarrow \perp$, $0x76 \rightarrow \sqsubseteq$, all `cmsy10`), so decoding the content stream against `cmsy10/cmr10/cmmi10` recovers the text character for character; a `pdftocairo` rendering confirms it independently. Consequences:

- **+ and $\otimes$ are different operators, and what this development has is $\otimes$.** `lem10_sum` and `lem17_sum` range over `CoalescedSum`, so they discharge the $\otimes$ conjuncts, not the + ones earlier drafts credited them with. + should be cheap: $D + E = D \perp \otimes E \perp$.
- **The strict arrow is two glyphs**, `openbullet` + `arrowright`; both $\rightarrow$ and $\circ\rightarrow$ extract as `!`, so every earlier extraction conflated two function spaces.
- **Lemma 17’s three powerdomain conjuncts $D_\natural$, $D_\#$, D_b were never stated**, though all three powerdomains have existed since r0029.
- **Theorem 12’s base theory is $T_\natural$ (natural)**, not an undecorated T : `pdftotext` renders $\natural/\#/\flat$ as `\[/\[/`, losing the accidental.
- **Two of Lemma 9’s conjuncts are misprints** — items 3 and 5 are *false as printed*, refuted on $D = E = \text{Prop}$, $F = \text{Prop} \times \text{Prop}$ by cardinality (10 vs 8, and 5 vs 3). The corrections are forced, and item 3’s is the universal property of $\otimes$ that the paper states three pages earlier. Recorded in `docs/StatementRecovery.md`; not yet put under the kernel.

Theorem 16’s second conjunct is false. The paper asserts the inclusion $i : \text{Fp}(D) \hookrightarrow (D \rightarrow D)$ is an embedding. Against a five-element bifinite domain with two incomparable minimal upper bounds — one that satisfies `thm16’s first conjunct`, so the refutation isolates the second — there are two incomparable maximal finitary projections below $f = \lambda x. m_1$ and no greatest one, while any monotone section would have to produce one. The sketch’s error is exact: $p \sqsubseteq f \iff \text{im}(p) \cap K(D) \subseteq S_f$, so what is needed is a normal set **contained in** S_f , whereas the paper takes the least normal set **containing** it. Those agree only when S_f is already normal. No choice of N_f repairs it. A positive statement remains available: the conjunct holds when every S_f has a greatest normal subposet, which bounded complete domains satisfy.

The idealSup repair, closing the third instance of one defect. `idealSup` now branches on `Order.IsIdeal (genIdeal S)` — the ideal *generated* by the union — rather than on the union itself being an ideal. `genIdeal_eq_sUnion_of_isIdeal` proves

the two agree wherever the old guard fired, so `lubOfDirected` and `mem_sSup_iff` kept their statements *and their proofs*, and Theorem 11 is byte-identical. `not_boundedComplete_{hoare,smyth,plotkin}` are retired, and `instBoundedCompleteHoare` (no bounded-completeness hypothesis needed) and `instBoundedCompleteSmyth` take their place. The witness is kept as three live lemmas rather than a docstring: the old guard still fails on it, and the repaired `sSup` returns the true least upper bound. A docstring cannot detect a regression.

Round r0029 ran four agents in parallel and closed the definition list.

#	Stream	Landed
1	<code>Powerdomain/Hoare.lean</code>	the Hoare (lower) powerdomain: Pf the finite non-empty subsets of $K(D)$, the lower pre-order, and <code>Domain</code> from Theorem 11
2	<code>Powerdomain/Smyth.lean</code>	the Smyth (upper) powerdomain, same shape, dual orientation
3	<code>Powerdomain/Plotkin.lean</code>	the Plotkin (convex) powerdomain under the Egli–Milner pre-order
4	<code>RecursiveDomain.lean</code>	the recursive domain equation, two formalizations of <i>universal domain</i> , and Theorem 21 — plus <code>recursiveDomain_funSpace</code> , the reflexive domain $D \cong (D \rightarrow D)$

There is no D_∞ to build. Earlier drafts of this inventory listed D_∞ (inverse limit) as an outstanding definition. Reading §7 directly refutes that: the section raises the chain $T_0 \rightarrow_{e_0} T_1 \rightarrow_{e_1} T_2 \rightarrow \dots$, says “This is all very informal, however; how are we to make this idea mathematically precise...?”, and answers with §7.1, *Solving domain equations with closures*. $D \cong D \rightarrow D$ is reached from Theorem 21 and Lemma 23, and the limit is taken inside $Fc(U)$. What stands in D_∞ ’s place is `Recursive.Solves / IsSolvable`, and the two formalizations of universal domain — `IsUniversal` (every domain of the class is a closure of U) and `IsUniversalRetract` (every one is a retract), which the paper states as two different sentences.

Pf is the finite non-empty subsets. The paper defines $Pf(S)$ that way and reserves $\bar{P}f(S)$ for the version including $\emptyset$; all three powerdomains are built over Pf . The distinction is load-bearing and in opposite directions for the three orderings: under the Hoare order $\emptyset \sqsubseteq v$ holds vacuously, so admitting $\emptyset$ would add a point strictly below $\{1\}$; under the Smyth order $\emptyset$ is a **top**, so it would add a spurious maximum; under Egli–Milner $\emptyset$ is comparable to nothing but itself, destroying `OrderBot`. The orchestrator’s brief said “finite subsets”; each agent read the PDF and corrected it.

Namespace per agent worked. Every r0029 declaration lives in `ScottDomains.{Hoare, Smyth, Plotkin, Recursive}`. Four agents, **zero** name collisions — against two in r0028, when five agents shared one namespace.

Round r0031 ran four agents and closed the definition list.

#	Stream	Landed
1	<code>ComputableFunction.lean</code>	the paper’s computable function , on Mathlib’s <code>REPred</code> — the last definition
2	<code>Powerdomain/BoundedComplete.lean</code>	Lemma 13 in the paper’s own wording, and a refutation: see the <code>idealSup</code> defect below
3	<code>Powerdomain/Universal.lean</code>	<code>isRepresentable_prod</code> — the product operator is representable over $P\ N$, the hypothesis Lemma 24 lacked ; plus $D \cong D \times D$
4	Theorem 18	still running at the time of writing

The `idealSup` defect — the same class, a third time. `IdealCompletion.idealSup` branches its dite on `Order.IsIdeal (⋃ ...)`. That is the membership condition for the *union*, but the union is the least upper bound only when the family is **directed**; for a merely **bounded** family the least upper bound is the ideal *generated* by the union. So `sSup` returns $\perp$ on a bounded non-directed family — kernel-checked in $P\ N$ with the incomparable compacts $\{0\}$ and $\{1\}$, bounded above by $\{0, 1\}$. This follows `ScottHom (r0006–r0011)` and `Smash (r0027)`, and the r0031 plan asserted the guard was correct, which was wrong. Consequently `not_boundedComplete_hoare`, `_smyth` and `_plotkin` are proved: **no `BoundedComplete` instance can be claimed for any ideal completion until `idealSup` is repaired**, by branching on the *generated* ideal rather than on the union being one.

Lemma 13 is nonetheless true and proved, in the paper’s own wording ($BddAbove\ S \rightarrow \exists\ I, IsLUB\ S\ I$, with the least element from `OrderBot`): D_b (Hoare) needs only the union, so it is bounded complete for **every** domain and the hypothesis is never consumed; $D_\#$ (Smyth) spends both hypotheses. There is no `lem13_plotkin` to prove — Lemma 13 names only $D]$ and $D[$, and Egli–Milner has no join for a bounded pair.

Two inventory rows below are corrected by r0031's reading of §7. Lemmas 28 and 30 are **not** about $P \sqsubseteq N$ and not about $F_c(U)$: both are stated for p -representability over $F_p(U)$, Lemma 28 over §7.3's ideal completion of dyadic half-open intervals and Lemma 30 over §7.4's bifinite universal domain. Lemma 23 is therefore **not** Lemma 28's function-space conjunct. Also, the paper proves $F(X) = X + X$ has **no** representation over $P \sqsubseteq N$, so $a +$ conjunct there is false, not missing.

Theorem 12 is recoverable after all. The row below says its “axioms T ” are never legible; pdftotext -layout extracts them cleanly — a continuous algebra $\otimes : E \times E \rightarrow E$ with associativity, commutativity and idempotence, where $T\#$ adds $s \otimes t \sqsubseteq s$ and $T\flat$ adds $s \sqsubseteq s \otimes t$. It belongs to §5.3 with Lemma 13, not to the §7 group.

Round r0028 ran five agents in parallel and roughly doubled the development: $27 \rightarrow 33$ modules, $4440 \rightarrow 8212$ lines, $199 \rightarrow 384$ theorems, $9 \rightarrow 15$ numbered results. Every new result was kernel-audited with `#print axioms`: all depend only on `propext`, `Classical.choice`, `Quot.sound`, and none on `sorryAx`.

#	r0028 stream	Landed
1	CoalescedSum.lean, Skeleton/Sum.lean	$D + E$ as a cpo (sumSup, sumCpo), then <code>lem10_sum</code> , <code>lem17_sum</code> , <code>lem17_smash</code> — Lemma 10 at 6 of 6 conjuncts, Lemma 17 at 5 of 5
2	IdealCompletion.lean	Theorem 11 , both halves, on Mathlib's <code>Order.Ideal</code> ; §5 unblocked $F_p(D)$ and $F_c(D)$ as posets, Theorem 16 (algebraic-lattice conjunct), Lemma 20 , and <code>IsClosure.domain_range</code> — Lemma 19 at the paper's strength
3	FinitaryProjectionPoset.lean, Skeleton/Section6b.lean	
4	MinimalUpperBounds.lean	
5	UniversalDomain.lean	minimal upper bounds, U, U^∞ , and <code>isPlotkinOrder_iff_mubClosure</code> — a characterization the paper does not state. Theorem 18 still open Theorem 22 and Lemma 23 , with <i>representable</i> defined from §7; opens the route to D^∞

A name clash lake build could not catch. Streams 3 and 5 each defined `IsClosure.apply_sSup_of_directed` and `isClosure_sSup`. The build passed at 971 jobs because no module imported both; the clash appeared the moment anything did — here, an axiom audit importing the pair. `isClosure_sSup` was the *same* statement proved twice; the single copy now lives in `Skeleton/Section6.lean` beside `IsClosure`, which both modules already import, so no call site changed. The two `apply_sSup_of_directeds` were *different* statements sharing a name — one indexed by a set of the subtype $\iota(im \ r)$, one by an ambient $D : Set \ \alpha$ with $D \subseteq im \ r$ — so the subtype form was renamed `IsClosure.apply_sSup_val_image_of_directed`.

A green build is not evidence that parallel work composes; importing every new module together is.

Row 5 counts lines matching `^(@[...] )?(theorem|lemma)` across the 37 modules.

The sorry burn-down (from r0026). The `sorry`s are deliberate scaffolding: fixed *statements* of outstanding results, confined to `ScottDomains/Skeleton/`, one file per agent worktree, so that three agents can prove them in parallel without any agent editing a declaration another depends on. Every other module remains `sorry`-free, and the count above is the burn-down metric — it goes $10 \rightarrow 0$. Round r0027, run as three agents in parallel, took it **10** $\rightarrow$ **1**.

#	Open statement	Result	State after r0027
1	prop15	Prop 15 — every bounded complete domain is bifinite	proved
2	thm18	Thm 18 — $D, D \rightarrow D$ domains $\implies D$ bifinite	sorry — the paper gives no proof, citing Smyth [Smy83a]; the obstacle is recorded in the docstring
3	lem19	Lem 19 — the image of a closure is a domain	proved , via <code>IsClosure.rangeCompletePartialOrder</code>
4–7	lem10_prod, lem10_smash, lem10_lift, lem10_strict	Lem 10 — bounded completeness closed under $\times, \otimes, ()^\perp, \rightarrow^\perp$. The $\rightarrow$ conjunct is already proved (Thm 7's bounded-complete half, r0007)	all four proved ; Lem 10 is then 5 of 6 conjuncts — $D + E$ is not stated
8–10	lem17_prod, lem17_lift, lem17_fun	Lem 17 — bifiniteness closed under $\times, ()^\perp, \rightarrow$	all three proved ; Lem 17 is then 3 of 5 conjuncts — $D \otimes E$ and $D + E$ are not stated

The + conjuncts, closed in r0028. CoalescedSum.lean was 181 lines of ingredients with no sSup and no cpo instance, so there was no $D + E$ to state a conjunct over. It now has both, and the guard is the membership condition `IsNonBotSum (sumCandidate (sumBase s))` — the defining predicate of the subtype — not directedness, so the defect fixed in ScottHom and then in Smash did not recur a third time. One wrinkle the smash did not have: $\alpha \otimes \beta$ carries no SupSet, so a summand must be selected first; that selection is not a second guard, because a set with an upper bound at all lies in one summand.

The smashSup defect (r0027). `lem10_smash` was not merely open: as `smashSup` stood, it was **false**, and the kernel confirmed a refutation. `smashSup` branched its dite on the base being nonempty *and directed*, so a merely **bounded** non-directed base fell to the adjoined $\perp$, which is not even an upper bound. Witness: $D = \text{Prop} \times \text{Prop}$, $E = \text{Prop}$, $s = \{\uparrow((\text{True}, \text{False}), \text{True}), \uparrow((\text{False}, \text{True}), \text{True})\}$, bounded by $\uparrow((\text{True}, \text{True}), \text{True})$. This is the same defect ScottHom.lean records having already hit for the function space. The repair branches on the coordinatewise supremum landing in `NonBotPair` — the condition under which it is an element of $D \otimes E$ at all, rather than a merely sufficient condition for it. `smashSup_of_directed` and `smashSup_of_empty` kept their statements and were reproved, which is the agreement claim stated in Lean rather than in prose, and `smashCpo` needed no change.

Kernel check on the r0027 merges (`#print axioms`): all ten proved statements plus `smashCpo`, `smashSup_of_directed` and `smashSup_of_empty` depend only on `propext`, `Classical.choice`, `Quot.sound` — `lem10_prod` and `lem19` do not even need `Classical.choice`. None depends on `sorryAx`. `thm18` does, as its `sorry` requires.

Row 2 counts only the paper’s 30 **numbered** results (Theorems / Lemmas / Proposition 1–30). Row 3 counts the claims the paper makes **in prose** rather than as numbered results; these are paper content too, and all twelve are formally verified:

#	Paper claim	Where	Lean
1	“the compact elements [of P N] are just the finite subsets of N ”	p. 9	<code>isCompactElement_iff_finite</code>
2	“ P N ... is a domain”	p. 9	<code>instance : Domain (Set X)</code>
3	“ $D \rightarrow E$ is a ... cpo”	Thm 7 proof	<code>instance : CompletePartialOrder (ScottHom α β)</code>
4	“... bounded complete ... whenever E is”	Thm 7 proof	<code>instance : BoundedComplete (ScottHom α β)</code>
5	“step(s) ... is continuous”	Thm 7 proof	<code>scottContinuous_stepFun</code>
6	“... and compact in the ordering on $D \rightarrow E$ ”	Thm 7 proof	<code>isCompactElement_step</code>
7	“they form a basis for $D \rightarrow E$ ”	Thm 7 proof	<code>instance : IsAlgebraic (ScottHom α β)</code>
8	every compact function is a <i>finite</i> join of step functions	Thm 7 proof, implicit	<code>exists_finite_isLUB_of_isCompactElement</code>
9	“an embedding is an injection”	§3.1	<code>IsEmbeddingProjectionPair.injective_embedding</code>
10	“a projection is a surjection”	§3.1	<code>IsEmbeddingProjectionPair.surjective_projection</code>
11	“it is easy to check that p_N ... is a finitary projection”	§3.1	<code>isFinitaryProjection_normalHom</code>
12	“the set of strict continuous functions $D \rightarrow E$ is also a cpo”	§2.1	<code>ScottDomains.strictHomCpo</code>

Six of those eleven are the body of **Theorem 7**, which is now **complete** — all four conjuncts of its conclusion (cpo, bounded complete, algebraic, countably based) are formally verified, as `ScottHom.isBoundedCompleteDomain_scottHom`.

It is proved under **weaker hypotheses than the paper states**: bounded completeness of D is never used. D need only be a domain. The function space is a cpo for any preordered D , algebraic when D and E are, bounded complete because E is, and countably based because D and E are.

The remaining theorems are supporting API: the $\ll$ calculus, the `compactsBelow` machinery, the pointwise order and suprema on $D \rightarrow E$, and the step-function adjunction. The paper assumes or elides all of it.

Reference audit (r0020). Of the theorems in the development, 16 are never cited elsewhere. Nine of those are *terminal by design* — they are the paper’s own claims, so nothing should cite them (`injective_embedding`, `surjective_projection`, the `isNormalIn_sUnion*` family and `mono_right` for Lemma 4, `singleton_bot_isNormalIn` for $\{\perp\} \in P(C)$, `isLeast_kleeneFix_le`, `eq_kleeneOperator_op`). The other six were speculative API written for callers that never appeared; they are **commented out in place**, each with a note on why it exists and what is instructive about it, and the build is unchanged — which also confirms that the three `@[simp]` ones among them were never firing implicitly.

The development is **45 modules, 14048 lines, 8 sorry, 0 other warnings** (measured by `scripts/counts.sh`). Seven of the eight sorrys are the newly *statable* Lemma 9 and Theorem 14 in `Skeleton/Recovered.lean`; the eighth is `thm18`. Counts of definitions, results and theorems are in the Progress table above — they are not repeated here, so that this section cannot drift out of step with it. What each round delivered:

#	Round	Module	Contents
1	r0003	<code>ScottDomains/WayBelow.lean</code>	way-below $\ll$ at $[\text{Preorder } \alpha]$, 7 theorems; $x \ll x \leftrightarrow$
2	r0004	<code>ScottDomains/Domain.lean</code>	<code>IsCompactElement</code> x holds by <code>Iff.rfl</code> <code>IsAlgebraic</code> , <code>Domain</code> (with the paper's countable-basis condition), <code>BoundedComplete</code> , 8 theorems, <code>Domain Prop</code>
3	r0005	<code>ScottDomains/Powerset.lean</code>	the paper's P N (p. 9): compacts of <code>Set X</code> are exactly the finite subsets, hence <code>Domain (Set N)</code> — the nondegenerate witness
4	r0006–r0007	<code>ScottDomains/ScottHom.lean</code>	the continuous function space $D \rightarrow E$: <code>ScottHom</code> , the pointwise order, <code>CompletePartialOrder</code> , and <code>BoundedComplete</code> when E is — Theorem 7's first sentence in full
5	r0008	<code>ScottDomains/StepFunction.lean</code>	the single step function <code>step k e</code> : continuity (from k compact), the adjunction $\text{step } k \ e \leq f \leftrightarrow e \leq f \ k$, and compactness in $D \rightarrow E$ (from e compact)
6	r0009	<code>ScottDomains/FunctionSpaceDomain.lean</code>	$D \rightarrow E$ is algebraic — the paper's “they form a basis for $D \rightarrow E$ ”; the two halves of <code>IsAlgebraic</code> use disjoint hypotheses (directedness needs only E bounded complete; the lub needs only D, E algebraic)
7	r0010	<code>ScottDomains/CompactFunction.lean</code>	every compact function is a finite join of step functions — the finiteness half of the basis claim
8	r0011	<code>ScottDomains/FunctionSpaceCountable.lean</code>	$K(D \rightarrow E)$ is countable, hence <code>Domain (ScottHom α β)</code> — Theorem 7 complete
9	r0012	<code>ScottDomains/NormalSubposet.lean</code>	the normal-subposet relation $\triangleleft$ and Lemma 4 , all four parts
10	r0012–r0013	<code>ScottDomains/Projection.lean</code>	embedding–projection pairs, projections, the paper's “an embedding is an injection, a projection is a surjection”;
11	r0014	<code>ScottDomains/FinitaryProjection.lean</code>	$\text{im}(p)$ is a cpo and finitary projections Lemma 5 — the compacts of $\text{im}(p)$ are $\text{im}(p) \cap K(D)$ (needs only that p is a projection), and $\text{im}(p) \cap K(D) \triangleleft K(D)$
12	r0015	<code>ScottDomains/NormalProjection.lean</code>	$p_N(x) = \bigsqcup \{y \in N \mid y \sqsubseteq x\}$: continuous, a projection, and $\text{im}(p_N) \cap K(D) = N$ — half of Theorem 6's correspondence, plus order preservation both ways
13	r0016	<code>ScottDomains/Theorem6.lean</code>	Theorem 6 — p_N is finitary, $p_N \circ \text{im}(p) \cap K(D) = p$, and the correspondence assembled

#	Round	Module	Contents
14	r0017	ScottDomains/FixedPoint.lean	Theorem 1 — $\bigsqcup_n f^n(\perp)$ is the least fixed point of a continuous f on a cpo. Not Mathlib reuse; see the §2 table
15	r0018	ScottDomains/UniformFixedPoint.lean	Theorem 3 — fix is the unique uniform fixed-point operator; $\downarrow a$ as a cpo
16	r0019	ScottDomains/Product.lean	$D \times E$ as a cpo (the one construction needing no case split) and Lemma 8 parts 1–3
17	r0021	ScottDomains/Currying.lean	Lemma 8.4 — currying, $D \rightarrow (E \rightarrow F) \equiv (D \times E) \rightarrow F$, and with it Lemma 8 complete
18	r0022	ScottDomains/EffectivePresentation.lean	§3.2’s effective presentation — the enumeration of the basis with its two decidability conditions
19	r0023	ScottDomains/Lift.lean	the lift $D\perp$ as a cpo, on Mathlib’s WithBot
20	r0024	ScottDomains/StrictHom.lean	the strict function space $D \rightarrow\perp E$ as a cpo — needs no case split, since both branches of ScottHom’s $s\text{Sup}$ are strict
21	r0025	ScottDomains/Smash.lean	§4.3’s smash product $D \otimes E$ — the non-bottom pairs with a new bottom adjoined, as a cpo
22	r0025	ScottDomains/Bifinite.lean	§6.1’s Plotkin order and bifinite domain

§2 and §3 are now complete — the only §3 omission is the paper’s *computable function*, which needs an r.e.-predicate notion Mathlib does not supply.

Dependency structure of the remaining definitions

#	Definition	Prerequisites	Status
1	smash product $D \otimes E$	Domain, OrderBot	✓ done, r0025
2	bifinite / Plotkin order	IsNormalIn ($\prec$, r0012)	✓ done, r0025
3	sum $D + E$	Domain	independent, ready
4	D_∞	embedding–projection pairs (r0012)	independent, but large (inverse limits)
5	the three powerdomains	the ideal completion (Theorem 11) — <i>not</i> built	blocked ; the three are independent of each other once it exists

Rows 1 and 2 were written as two modules and checked in a single build, which is the parallelism the structure actually admits — the bottleneck is the build-and-fix loop, not the writing.

Next: the sum $D + E$, then Theorem 11 to unblock the powerdomains, then Lemma 9 (the strict analogues of Lemma 8) and Lemma 10 (closure of bounded completeness).

Work counts

- **Reuse from Mathlib — no work (11):** poset, directed, cpo, $\perp$, monotone, continuous, fixed-point operator, Schröder–Bernstein (2), algebraic lattice, product, λ -notation. (Theorem 1 was listed here in an earlier draft and has been removed: Mathlib’s `OrderHom.lfp` is Knaster–Tarski over a complete lattice, not Kleene’s $\bigsqcup_n f^n(\perp)$ over a cpo. It is proved in `ScottDomains/FixedPoint.lean`, so **29** numbered results needed proof, not 28.)
- **Generalize / adapt (4):** compact element (`IsCompactElement` — its *definition* is already stated at `[PartialOrder α]` and so applies to a dcpo verbatim; what is `CompleteLattice`-only is every lemma about it, so the dcpo API must be re-proved), sum, lift (`WithBot/Sum`), ideal completion (`Order.Ideal`).
- **Definitions to define — new (≈ 13 ; 4 done in r0003–r0004, 9 remaining):** way-below $\ll$ (**done** — `ScottDomains/WayBelow.lean`), algebraic cpo, **domain**, bounded-complete (**all three done** — `ScottDomains/Domain.lean`), embedding–projection

pair, (finitary) projection, normal subposet, effective presentation, smash product, the three powerdomains (Hoare / Smyth / Plotkin), bifinite / Plotkin order, and D^∞ .

- **Theorems to prove (28):** 28 of the paper's 30 numbered results — Theorems 3, 6, 7, 11, 12, 14, 16, 18, 21, 22, 25, 26, 27, 29; Lemmas 4, 5, 8–10, 13, 17, 19, 20, 23, 24, 28, 30; Proposition 15. Only Theorems 1 & 2 come free from Mathlib.

Bottom line: ≈ 13 definitions to define + 28 results to prove, on top of 12 reused Mathlib foundations. After r0004: 9 definitions + 28 results remain.

Lean column legend: $\checkmark$ reuse Mathlib (name given) $\cdot$ $\sim$ partial (Mathlib has a related or lattice-only version — generalize) $\cdot$ $\times$ define / prove (absent from Mathlib v4.32.2, confirmed by grep).

Statements are paraphrased/de-garbled from the PDF (1990 Type-3 fonts render $\rightarrow$ as $!$, drop f i ligatures, mangle math), so read them as a guide to *what* each result is. Notation: $\sqsubseteq$ order, $\bigsqcup$ directed sup, $\perp$ bottom, $\ll$ way-below, $K(D)$ compacts, $Fp(D)$ finitary projections.

§2 Recursive definitions of functions

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
2.1	Def	—	poset (partially ordered set)	$\checkmark$ PartialOrder
2.1	Def	—	directed subset M : every finite $u \subseteq M$ has an upper bound in M	$\checkmark$ DirectedOn / IsDirected
2.1	Def	—	cpo : poset in which every directed M has a lub $\bigsqcup M$	$\checkmark$ CompletePartialOrder (chains: Ω CompletePartialOrder)
2.1	Def	—	bottom $\perp$ (least element)	$\checkmark$ OrderBot
2.1	Def	—	monotone function	$\checkmark$ Monotone / OrderHom
2.1	Def	—	continuous : monotone, $f(\bigsqcup M) = \bigsqcup f(M)$ for directed M	$\checkmark$ ScottContinuous
2.1	Thm	1	Fixed-Point Theorem: $f : D \rightarrow D$ continuous $\implies$ least fixed point $\bigsqcup_n f^n(\perp)$	$\checkmark$ proved (r0017) — ScottDomains.theorem1. Not Mathlib reuse: OrderHom.lfp is Knaster–Tarski over a <i>complete lattice</i> with only monotonicity, a different theorem
2.2	Thm	2	Schröder–Bernstein for sets	$\checkmark$ Function.schroeder_bernstein (SetTheory/Cardinal/SchroederBernstein.lean:90) — the name in an earlier draft of this row, Function.Embedding.schroederBernstein, does not exist
2.3	Def	—	fixed-point operator (uniform)	$\checkmark$ OrderHom.lfp / LawfulFix
2.3	Thm	3	The standard operator is the unique uniform fixed-point operator	$\checkmark$ proved (r0018) — ScottDomains.theorem3; a fixed point operator is formalized as a family over every CompletePartialOrder in a universe

§3 Effectively presented domains

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
3.1	Def	—	compact element x : $x \sqsubseteq \bigsqcup M$ (dir.) $\implies x \sqsubseteq y$ some $y \in M$; $K(D)$	$\sim$ IsCompactElement — def is [PartialOrder], so reusable on a dcpo; its lemmas are all CompleteLattice-only
3.1	Def	—	way-below $\ll$ (approximation relation behind compactness)	$\checkmark$ ScottDomains.WayBelow (r0003) — $x \ll x \leftrightarrow$ IsCompactElement x by Iff.rfl
3.1	Def	—	algebraic cpo: $x = \bigsqcup \{x' \in K(D) : x' \sqsubseteq x\}$ (directed)	$\checkmark$ ScottDomains.IsAlgebraic (r0004)
3.1	Def	—	domain = algebraic cpo whose basis $K(D)$ is countable (the paper's definition, p. 9 — the countability condition was missing from an earlier draft of this row)	$\checkmark$ ScottDomains.Domain (r0004)

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
3.1	Def	—	bounded complete: $\perp +$ every bounded subset has a sup — a <i>separate</i> predicate; the paper composes them as “bounded complete domain” (Thm 7, Lem 10, Lem 13, Thm 14), which is the literature’s <i>Scott domain</i>	✓ <code>ScottDomains.BoundedComplete</code> (r0004); the compound is <code>[Domain α] [BoundedComplete α]</code>
3.1	Def	—	(countably based) algebraic lattice	✓ <code>IsCompactlyGenerated (+CompleteLattice)</code>
3.1	Def	—	embedding–projection pair (g, f)	✓ <code>ScottHom.IsEmbeddingProjectionPair</code> (r0012)
3.1	Def	—	projection; finitary projection $p: p \circ p = p \sqsubseteq \text{id}$, $\text{im}(p)$ a domain	✓ <code>ScottHom.IsProjection</code> (r0012), <code>ScottHom.IsFinitaryProjection</code> (r0013) — $\text{im}(p)$ carries a <code>CompletePartialOrder</code> via <code>IsProjection.rangeCompletePartialOrder</code>
3.1	Def	—	normal subposet / substructure	✓ <code>ScottDomains.IsNormalIn</code> , notation $\triangleleft$ (r0012)
3.1	Lem	4	$\langle P(C), \triangleleft \rangle$ of substructures is a cpo with $\{\perp\}$ least	✓ proved (r0012), all four parts
3.1	Lem	5	p finitary projection $\implies$ compacts of $\text{im}(p)$ are $\text{im}(p) \cap K(D)$, and $\text{im}(p) \cap K(D) \triangleleft K(D)$	✓ proved (r0014) — <code>IsProjection.isCompactElement_iff</code> (needs only <i>projection</i>) and <code>IsFinitaryProjection.isNormalIn_compacts</code>
3.1	Thm	6	Isomorphism: normal substructures $\cong \text{Fp}(D)$ (finitary projections)	✓ proved (r0015–r0016) — <code>ScottDomains.theorem6</code>
3.1	Thm	7	D, E bounded-complete domains $\implies D \rightarrow E$ bounded-complete domain	✓ proved (r0006–r0011) — <code>ScottHom.isBoundedCompleteDomain_scottHom</code> ; D bounded complete is not needed
3.2	Def	—	effective presentation $d: \mathbb{N} \rightarrow K(D)$; effectively presented domain	✓ <code>ScottDomains.EffectivePresentation</code> (r0022). The paper’s computable function is ✓ defined (r0031) in <code>ComputableFunction.lean</code> , on Mathlib’s REPred (Mathlib/Computability/RE.lean:157). An earlier draft of this row said “Mathlib v4.32.2 has no RePred or equivalent (grep finds none)” — that grep used the wrong capitalization, and the claim was false. Two caveats recorded there: <code>decidableLE</code> is too weak to prove anything computable (a <code>Decidable</code> instance may be <code>Classical.dec</code>), so results need an explicit <code>RecursiveLE</code> hypothesis; and composition awaits REPred closure under $\wedge$ and $\exists$, which Mathlib’s five-lemma API does not supply

§4 Operators and functions

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
4.1	Def	—	product $D \times E$	✓ Prod order from Mathlib; the cpo instance is <code>ScottDomains.instCompletePartialOrderProd</code> (r0019) — Mathlib has <code>Prod.supSet</code> and <code>isLUB_prod</code> but no cpo instance
4.2	Def	—	Church’s λ-notation (continuous abstraction)	✓ <code>OrderHom / ωCPO ContinuousHom</code>
4.3	Def	—	smash product $D \otimes E$	✓ <code>ScottDomains.Smash + smashCpo</code> (r0025)
4.4	Def	—	sum $D + E$; lift $D \perp$	both ✓ — lift <code>ScottDomains.liftCpo</code> on <code>WithBot</code> (r0023); the coalesced sum <code>CoalescedSum</code> with <code>sumSup</code> and <code>sumCpo</code> (r0028), its <code>sSup</code> guarded on landing in <code>NonBotSum</code>

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
4.x	Lem	8	$D \times E \cong E \times D$; $(D \times E) \times F \cong D \times (E \times F)$; $D \rightarrow (E \times F) \cong (D \rightarrow E) \times (D \rightarrow F)$; $D \rightarrow (E \rightarrow F) \cong (D \times E) \rightarrow F$	✓ proved — <code>prodComm</code> , <code>prodAssoc</code> , <code>scottHomProd</code> (r0019); <code>scottHomCurry</code> (r0021)
4.x	Lem	9	Iso laws over D, E, F : $D \otimes E \cong E \otimes D$; $(D \otimes E) \otimes F \cong D \otimes (E \otimes F)$; $(E \otimes F) \multimap D \cong (E \multimap D) \times (F \multimap D)$; $D \multimap (E \multimap F) \cong (D \otimes E) \multimap F$; $D \otimes (E \otimes F) \cong (D \otimes E) \otimes (D \otimes F)$; $D \perp \multimap E \cong D \rightarrow E$	✓ proved (r0034) — six named <code>=o</code> under <code>ScottDomains.Isomorphism</code> : <code>smashComm</code> , <code>smashAssoc</code> , <code>coalescedSumCoproduct</code> , <code>smashCurry</code> , <code>smashDistribCoalescedSum</code> , <code>liftStrictHomIso</code> . Items 3 and 5 are false as printed and are kernel-checked negations, <code>lem9_3_printed_false</code> and <code>lem9_5_printed_false</code> in <code>Isomorphism/Counterexample.lean</code> . The witnesses are not the cardinality argument of <code>docs/StatementRecovery.md</code> ($D = E = \text{Prop}$, $F = \text{Prop} \times \text{Prop}$), which needs <code>Fintype</code> instances for <code>WithBot</code> of a subtype plus an enumeration of the strict monotone maps: they are <code>PUnit</code> -based ($D = \text{PUnit}$, $E = F = \text{Prop}$ for item 3; $D = \text{Prop}$, $E = \text{PUnit}$, $F = \text{Prop}$ for item 5) and separate the sides by the coarsest invariant an order isomorphism must preserve — one element versus more than one — discharged by <code>Equiv.subsingleton</code> . Each witness is chosen so the <i>corrected</i> law survives it, so the separation is specific to the misprint. $\multimap$ is the strict function space; it and $\rightarrow$ both extract as !
4.5	Lem	10	D, E bounded complete $\implies \rightarrow, \multimap, \times, \otimes, \oplus, +, () \perp$ bounded complete	✓ 7 of 7 (r0034) — $\rightarrow$ r0007; $\times, \otimes, () \perp, \multimap$ r0027 (<code>Skeleton/Lemma10.lean</code>); $\oplus$ r0028 (<code>Skeleton/Sum.lean</code>); $+$ as <code>ClosureProperties.lem10_separated</code> , via $D + E = D \perp \oplus E \perp$ — §4.4’s own definition, $+$ being a different operator from $\oplus$. Now also stated as one theorem, <code>ClosureProperties.lemma10</code> , a conjunction over the paper’s operator list, so the conjunct count is kernel-checked rather than prose. The docstring paraphrase in <code>Skeleton/Lemma10.lean</code> is six wide because $\oplus$ prints blank under <code>pdftotext</code> — a fourth instance of the glyph-dropping mechanism
4.5	Thm	11	Ideal completion of a countable pre-order is a domain (all domains so arise)	✓ proved (r0028) — <code>IdealCompletion.thm11</code> and <code>thm11_converse</code> , on Mathlib’s <code>Order.Ideal</code>
4.5	Thm	12	Initiality of a continuous algebra satisfying the axioms $\mathbf{T}_\natural$ (natural — not an undecorated $\mathbf{T}$; <code>pdftotext</code> renders $\natural/\#/\flat$ as $\backslash/[/[)$	✓ proved (r0032) — <code>ContinuousAlgebra.thm12_plotkin</code> , <code>thm12_smyth</code> , <code>thm12_hoare</code> : $\mathbf{T}_\natural \rightarrow \mathbf{D}_\natural$, $\mathbf{T}_\# \rightarrow \mathbf{D}_\#$, $\mathbf{T}_\flat \rightarrow \mathbf{D}_\flat$, each $\exists!$ h , existence <i>and</i> uniqueness, factoring through $\{ \cdot \}$ rather than the weaker principal ideals. $[\text{IsAlgebraic } \mathbf{D}]$ is the whole hypothesis — countability of $K(\mathbf{D})$ is never used, and bounded completeness is never needed. Each free algebra is also proved a model of its own theory, a gap the plan had not named
4.5	Lem	13	D bounded complete $\implies \text{powerdomains } D], D[$ bounded complete	✓ proved (r0031) — <code>PowerdomainBC.lem13_hoare</code> , <code>lem13_smyth</code> , in the paper’s wording. $\mathbf{D}_\flat$ needs no bounded-completeness hypothesis at all; there is no convex conjunct to prove

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
4.5	Thm	14	Plotkin’s SFP characterization: D bifinite $\iff D$ a domain and $K(D)$ a Plotkin order (<i>not</i> “equivalent characterizations of an (algebraic/BC) domain” — that reading of the row was wrong; the theorem is two items about bifiniteness)	✓ proved (r0036) — <code>Recovered.thm14</code> , both directions, from <code>SFP.thm14_forward</code> and <code>SFP.thm14_converse</code> in <code>ScottDomains/SFP.lean</code> . Of the four gaps r0034 recorded in <code>thm14’s</code> <code>docstring</code> , two were real and two were false constraints. Real: gap 2, the bridge <code>Set.range ↑(toFp hN) = N</code> for finite normal N , proved as <code>SFP.range_normalHom_of_finite</code> — for finite N the directed set $N \cap \downarrow x$ contains its own greatest element, which is $p_N(x)$; for infinite N the statement is false. Gap 4, the two finite-combinatorial lemmas, proved as <code>SFP.exists_upperBound_of_finite_subset</code> and <code>SFP.exists_greatest_of_finite</code> . False constraints: gap 1 (the <code>FpLattice</code> section sits at <code>[Domain α]</code>) never binds, because on a finite image $\text{im}(p) \cap K(D) = \text{im}(p)$, so the forward direction runs entirely in D and touches no <code>Fp(D)</code> machinery; gap 3 (<code>IsLUB</code> in $\uparrow(Fp \alpha)$ weaker than in <code>ScottHom α α</code>) never arises, because leastness of <code>id</code> is proved in the function space directly from algebraicity. One ingredient the r0034 note omitted is needed: <code>M.Nonempty</code> , since <code>IsCompactElement</code> quantifies over nonempty directed sets and Mathlib’s <code>DirectedOn</code> is vacuous on $\emptyset$ — supplied by <code>SFP.isFinitaryProjection_const_bot</code>

§5 Powerdomains

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
5.1	Def	—	powerdomain (non-deterministic outcomes)	✓ each of the three is <code>IdealCompletion (Pf K(D))</code> under its pre-order (r0029)
5.2	Def	—	Hoare (lower), Smyth (upper), Plotkin (convex) powerdomains	✓ <code>ScottDomains.Hoare.Powerdomain</code> , <code>Smyth.Powerdomain</code> , <code>Plotkin.Powerdomain</code> (r0029), each with its <code>Domain</code> instance from Theorem 11 and its compacts characterized as the principal ideals
5.3	—	—	Universal & closure properties (see Lem 13, 28, 30)	✗ prove — r0030 wave A

§6 Bifinite (SFP) domains

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
6.1	Def	—	Plotkin order / bifinite (SFP) domain	✓ <code>ScottDomains.IsPlotkinOrder</code> , <code>IsBifinite</code> (r0025)
6.1	Prop	15	Every bounded-complete domain is bifinite	✓ proved (r0027) — <code>ScottDomains.prop15</code> , the paper’s own proof over <code>lubClosure u</code>
6.1	Thm	16	D bifinite $\implies Fp(D)$ is an algebraic lattice	settled in all three directions as of r0034: the algebraic-lattice conjunct ✓ proved (r0028) as <code>ScottDomains.thm16</code> ; the $Fp(D) \hookrightarrow (D \rightarrow D)$ embedding conjunct ✗ refuted (r0032) in <code>FinitaryProjectionEmbedding.lean</code> , kernel-checked, with the sketch’s exact error identified; and the conjunct ✓ holds under a named hypothesis — <code>Section62.thm16_positive</code> with <code>HasGreatestStableNormal</code> , plus <code>thm16_positive_isEmbeddingProjectionPair</code> giving the pair itself (r0034). Bounded complete domains satisfy the hypothesis

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
6.2	Lem	17	D, E bifinite $\implies \rightarrow, \circ\rightarrow, \times, \otimes, \oplus, +, ()\perp, D^\sharp, D^\#, D^\flat$ bifinite	✓ 10 of 10 (r0034), with one qualification — $\times, ()\perp, \rightarrow$ r0027; $\otimes, \oplus$ r0028; and r0034 adds <code>lem17_separated</code> ($+$), <code>lem17_strictFun</code> ($\circ\rightarrow$) and <code>lem17_hoare/lem17_smyth/lem17_plotkin</code> for $D^\flat/D^\#/D^\sharp$. The three powerdomain conjuncts had been dropped from the extraction with their glyphs, not for want of the objects — all three powerdomains have existed since r0029. One lemma covers all three: the obvious Hoare-maximal candidate $\{n \in \mathbb{N} \mid \exists y \in w, n \sqsubseteq y\}$ is greatest for $\sqsubseteq^\flat$ but its Smyth conjunct fails for $\sqsubseteq^\sharp$; the image $p_N[w]$ is greatest for all three (<code>isNormalIn_image_principal</code> over <code>SelectsGreatest</code>). Now stated as one theorem, <code>ClosureProperties.lemma17</code> . Qualification: the $\rightarrow$ and $\circ\rightarrow$ conjuncts carry <code>[BoundedComplete β]</code> , inherited from the step-function decomposition — stronger than the paper states, and §6 exists precisely to avoid bounded completeness. Removing it is a real open item
6.2	Thm	18	If D and $D \rightarrow D$ are domains, then D is bifinite	✗ prove — the development’s only remaining sorry, and no longer blocked on an unobtainable source. [Smy83a] is needed for attribution only: r0034 recovered a complete proof of the same statement from Jung 1989, which is on disk , and mapped it to five steps in Section62.lean. Step 5 is <code>isBifinite_iff_mubClosure</code> (r0028). r0036 proved steps 2 and 3 in <code>JungSFP.lean</code> : <code>lemma213</code> and <code>thm214</code> , the bifurcation into bifinite-or-(vii), and <code>lemma217</code> , the cardinality argument, spending countability exactly once via <code>Function.cantor_surjective</code> against <code>Domain.countable_compacts</code> . r0037 proved step 4 and the assembly (<code>JungFinite.lemma129</code> , <code>lemma22</code> , <code>thm18_of_propertyM</code>) and the consequence of step 1 (<code>JungNets</code>), leaving exactly two named propositions : Jung’s Theorem 1.37 — which reads “a dcpo with continuous function space is bicomplete ”, <i>not</i> “has property m”; the latter is a separate inference inside the proof of Theorem 2.3 that Jung never proves and that <code>JungNets.exists_minimal_upperBounds_le</code> now supplies by Zorn downwards — and his Corollary 1.36 . The join is kernel-checked by <code>scripts/check-thm18-composition.sh</code> , which elaborates the two agents’ results in one environment. Mathlib has no Iwamura lemma (0 hits for Iwamura Markowsky), which Jung’s Corollary 1.3 rests on, so Theorem 1.37 is blocked on a missing theorem rather than missing notation. What unblocked three failed rounds was a missing bridge, not a tactic: the development states minimal upper bounds relative to $K(D)$, while Jung applies minimality to bounds not known to be compact — <code>isCompactElement_of_minimal_upperBounds</code> (his Prop 1.9 for <code>IsAlgebraic</code>) closes the gap, and without it f_A ’s monotonicity is false. IsLDomain is not needed: Lemma 2.17 uses only condition (vii) of Jung’s Theorem 2.10, so <code>thm214</code> ’s second disjunct is (vii)-minus-existence, not the literature’s “algebraic L-domain”. Remaining: Jung’s Lemma 1.29 (property M for pairs $\implies$ for all finite sets, the cheapest and the one that makes <code>lemma217</code> directly consumable), step 4 (Lemma 2.2, needing Rado’s Selection Theorem and Corollary 1.36), and step 1 (Theorem 1.37, a separate development over ordinal-indexed codirected nets). r0031’s (★) is equivalent to Theorem 18, not below it, and Jung’s proof never passes through it
6.2	Lem	19	closure $r:D \rightarrow D$ ($r \circ r = r \sqcap \text{id}$) $\implies \text{im}(r)$ is a domain	✓ proved — <code>lem19</code> (r0027) gives the cpo structure; <code>IsClosure.domain_range</code> (r0028) gives the paper’s full strength, $\text{im}(r)$ a domain with basis $\{r(k) \mid k \in K(D)\}$
6.2	Lem	20	D domain $\implies Fc(D)$ (finitary closures) is a cpo	✓ proved (r0028) — <code>ScottDomains.lem20</code> , over $Fc \ \alpha$ with the pointwise order

§7 Recursive definitions of domains (universal domain, D_∞)

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
7	Def	—	recursive domain equation; universal domain. (Earlier drafts of this row said “ D_∞ (inverse limit)” — §7 builds no inverse limit; see the note under Progress)	✓ <code>Recursive.Solves / IsSolvable</code> , and <code>Recursive.IsUniversal / IsUniversalRetract</code> for the paper’s two phrasings (r0029)
7	Thm	21	F representable over $\text{cpo } U \implies$ a domain D with $D \cong F(D)$	✓ proved (r0029) — <code>ScottDomains.Recursive.thm21</code> ; with <code>IsRepresentable₂.diag</code> and Lemma 23 it yields <code>recursiveDomain_funSpace</code> , the reflexive domain $D \cong (D \rightarrow D)$
7	Thm	22	Any countably-based algebraic lattice L : a closure $r : P(\mathbb{N}) \rightarrow L$	✓ proved (r0028) — <code>ScottDomains.thm22</code> , with <code>thm22_of_isCompactlyGenerated</code> the Mathlib-vocabulary form
7	Lem	23	The function-space operator is representable over $P(\mathbb{N})$	✓ proved (r0028) — <code>ScottDomains.lem23</code>
7	Lem	24	$U \text{ cpo}; \times \text{ and } \rightarrow$ representable over $U \implies$ (setup for universality)	✓ proved (r0032) — <code>Universality.lem24</code> , concluding also that D, E are closures of U , which its own proof needs
7	Thm	25	U non-trivial domain representing $\times, \rightarrow \implies U$ universal	✓ proved (r0032) — <code>Universality.thm25</code> , with <code>thm25_powerset</code> instantiating it at $P \ N$ and <code>thm25_isUniversal</code> in <code>Recursive.IsUniversal</code> form. Proved at cpo strength: no step spends algebraicity or countability, so it is stronger than the paper’s statement
7	Thm	26	Any signature $(s_1, \dots, s_n)$: combinations $F_1, \dots, F_n$ over $S, K, \text{fst}, \text{snd}$ into which every continuous algebra carried by a retract of D embeds as a subalgebra. An earlier draft of this row said “combinators solving the equations” — §7.2 solves nothing; it is an Engeler-style universal-algebra result and <code>Recursive.Solves</code> is the wrong vocabulary	✓ proved (r0034) — <code>Combinator.thm26</code> , with <code>thm26_subalgebra</code> , <code>thm26_retract</code> , <code>exists_lambdaModel_of_thm25</code> , and <code>Comb/combEval</code> so the F_i are genuinely combinations. Signature indexed $\text{Fin } n \rightarrow \mathbb{N}$, because F_i reads its slot by applying <code>snd</code> exactly i times. Carries $hs : \forall i, 0 < s_i$: the theorem is <i>false</i> for a signature admitting arity 0, which the paper explicitly allows — two one-point retracts with different constants must embed onto the same F_i , and <code>fst($\psi(x)$) = x</code> forces them equal. That argument is prose in the docstring, not kernel-checked

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
7	Thm	27	Any bounded-complete D : a projection of the universal domain onto D	✓ proved, unconditionally (r0036) — <code>Atomless.thm27</code>, with no hypothesis beyond <code>[Domain D] [BoundedComplete D]</code>, and non-vacuity compiled in <code>(example := thm27 Dyadic.U)</code>. Neither Vaught’s theorem nor a Boolean algebra is used, correcting two claims this file previously made. <code>IsNormallyRepresented</code> $\mathfrak{r}$(<code>compacts D</code>) is discharged by <code>Atomless.isNormallyRepresented</code>: what the theorem consumes is three properties of a map $\psi : K(D) \rightarrow U_0$ — order embedding into the superset order, $\psi \perp = [\emptyset, 1]$, and $\psi a \cap \psi b$ empty when $\{a, b\}$ is unbounded and $\psi(a \sqcup b)$ when bounded — and the third is also what makes <code>range</code> ψ normal, so the paper’s unproved final clause is four lines. ψ reads points of S as binary digit sequences and adjoins <code>enum n</code> to a branch only when <code>Legal</code>, which demands boundedness <i>and</i> that the join drag in no earlier <code>enum i</code> the branch does not carry; that second conjunct is what keeps ψ a finite union of intervals. Same mathematics as the paper’s atom splitting, stated on one element rather than on the atoms of a finite subalgebra. The Mathlib measurement stands — v4.32.2 has zero occurrences of <code>IsAtomless</code> — so the paper’s own route would have had to be built from nothing. The conditional form is retained as <code>Dyadic.thm27</code>, which builds $e : \text{ScottHom } D \ U, p : \text{ScottHom } U \ D$ with $p \circ e = \text{id}, e \circ p \sqsubseteq \text{id}$, going directly between D and U rather than through <code>normalHom/im(p)</code>. Normality is spent in exactly one place, making $\{k \mid \psi k \in I\}$ directed; <code>BoundedComplete D</code> is consumed inside the hypothesis. The open step is <code>IsNormallyRepresented</code>, which needs Vaught’s theorem — the countable atomless Boolean algebra is unique up to isomorphism. Mathlib v4.32.2 has zero occurrences of <code>IsAtomless</code> and none in <code>ModelTheory/</code>. Two smaller steps also remain: that $B = U_0 \cup \{\emptyset\}$ is a Boolean algebra (<code>isBasic_inter</code> is the intersection half, proved), and that the embedding cuts down to a <i>normal</i> subposet, which the paper asserts without proof. <code>prop15</code> and <code>thm22</code> are not on this route — §7.3 cites no earlier result of the paper

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
7	Lem	28	Operators $\rightarrow, \multimap, \times, \otimes, +, \oplus, ()^\perp, ()^\sharp, ()^\flat$ — nine, read off a 600 dpi rendering of printed page 42 in r0036, since pdftotext emits <code>!, !, , ; +, ; ()?, ()], ()[</code> for the whole list. Two earlier drafts of this row were wrong in opposite directions: seven operators, then nine including $()^\flat$ and omitting $\multimap$. $()^\flat$ is not here — §7.4 opens by saying it cannot be representable over U, as it does not preserve bounded completeness. All nine are p-representable over $F_p(U)$, <i>not</i> the closure notion of <code>IsRepresentable over $F_c(U)$</code>	~ 7 of 9 over the paper’s own Dyadic U, with no hypothesis (r0037): $\rightarrow, \multimap, \times, \otimes, +, \oplus, ()^\perp$. <code>PRepSum.pairAtU</code> derives the paper’s retraction pair at U from the now-unconditional <code>Atomless.thm27</code> in four lines; <code>PRepFun</code> proves $\rightarrow, \multimap, \otimes$ conditionally on that pair and supplies the two <code>Domain</code> instances the development lacked (<code>strictHomDomain</code>, <code>smashDomain</code>); <code>PRepSum</code> proves $+$, $\oplus$ and closes the algebraicity gap (<code>isAlgebraic_coalescedSum</code>); <code>Lemma28AtU.lean</code> is the orchestrator’s join, lifting the first three to U. The per-conjunct cost of lifting is Theorem 27’s hypothesis that the operator’s <i>result</i> be a bounded complete domain — which is Lemma 10 — so Lemma 10 and Lemma 28 compose. <code>lemma28AtU_of'</code> has arity 2, against 9 for <code>PRep.lemma28_of</code>. $()^\sharp$ and $()^\flat$ remain, and their obstruction is not the definability one earlier rounds recorded (<code>smyth0p/hoare0p</code> are definable on <code>Cpo</code>, r0036): the development defines no action of a map on either powerdomain, so there is no $r \mapsto r^\sharp$ to build the conjugating family from; the natural construction wants $p(K(D)) \subseteq K(D)$ for a finitary projection, which no round has settled. Earlier position, retained for the record: 2 of 9 at the paper’s own notion (r0036) — <code>PRep.rep_prod($\times$)</code> and <code>PRep.rep_lift($()^\perp$)</code> at <code>IsPRepresentable</code> over <code>Fp U</code>, with <code>PRep.Lemma28</code> the nine-fold conjunction and <code>lemma28_of</code> taking nine named hypotheses, so the count is kernel-checked rather than prose. This is a <i>smaller</i> number than the 3 of 9 r0034 recorded and a better position: those three do not transfer, and that is now kernel-checked. <code>Fp</code> adds an obligation <code>Fc</code> never had (<code>IsFinitaryProjection p</code> carries <code>Domain</code> $\mathfrak{r}(\text{Set.range } p)$), and the hypothesis is incompatible — <code>Combinator.Rettracts</code> gives $\text{id} \sqsubseteq \text{gr} \circ \text{fn}$ where the projection scheme needs $\text{gr} \circ \text{fn} \sqsubseteq \text{id}$, and <code>gr_fn_eq_of_both</code> proves holding both forces $U \cong V$. $\otimes$ and $\oplus$ are no longer refuted: a projection has $p \perp = \perp$, so r0034’s three-chain counterexample does not apply. <code>isFinitaryProjection_sSup</code> is the keystone for the remaining seven — over a domain the directed supremum of finitary projections is finitary, so least upper bounds in <code>Fp(D)</code> are pointwise, which every conjunct’s continuity proof needs; each is then 120–180 lines. Two docstring claims corrected: <code>PRepresentable.lean</code> said Lemma 28 is the <code>Fc</code> notion (§7.3’s “for the remainder of this section” refutes it), and <code>CombinatorRep.lean</code> claimed $()^\sharp/()^\flat$ are undefinable on <code>Cpo</code> — <code>smyth0p/hoare0p</code> compile, the <code>[Domain D]</code> being spent only on countability of the result

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
7	Thm	29	D bifinite $\implies D^+$ bifinite; and solving $D \cong D^+$ — record as split , the two sentences have different evidential status	~ first sentence proved (r0034) — <code>BifiniteUniversal.thm29</code>, with <code>Plus D = IdealCompletion (MPair ι (compact D))</code>. Two kernel-checked defects in §7.4: the printed relation is <i>not reflexive</i> ($b = (\perp, \emptyset)$ fails $b \vdash b$, as does $(\text{false}, \{\text{true}\})$ over <code>Bool</code>), so it is the strict part and the order is its reflexive closure — exactly the identification §7.4 then performs by hand; and the worked example reverses its own definition ($b \vdash a$ where both papers give $a \vdash b$). A rival repair (Smyth order on covers) also contains the printed relation and agrees at stages 0–2; the paper’s own element counts select between them — 1, 2, 5, 20 for the adopted reading against 1, 2, 5, 21 for Smyth. V is built (r0036) — <code>Colimit.V</code>, with <code>domain_V</code>, <code>isBifinite_V</code> and <code>isoPlus : V $\approx$ o Plus V</code>, so $D \cong D^+$ is solved. <code>Recursive.thm21</code> indeed does not apply, $D \mapsto D^+$ being a functor on domains rather than an operator representable over a fixed cpo; the colimit is taken at the level of countable posets and <code>IdealCompletion.thm11</code> is applied once at the end, so V needs no cpo construction. Shape: the Antisymmetrization of a pre-order on Σn, <code>Stg n</code> — forced, since <code>MPair (MPair A)</code> does not typecheck (<code>MPair</code> is a pre-order) — with <code>Nat.leRecOn</code> giving transport with zero casts, so the predicted dependent-transport cost did not materialize. A third printed defect in §7.4, kernel-checked as <code>Colimit.stgEmb_ne_mk_eta</code>: the paper (and <code>BifiniteUniversal.lean</code>) say the colimit is along η, $x \mapsto (x, \{x\})$, and that colimit is not a fixed point of M — reading $(x, u) \in M(A_N)$ one stage later gives base $[(x, \{x\})]$ where the colimit map needs $[(x, u)]$, and they agree only when $\uparrow u = \uparrow x$, failing at §7.4’s own $b = (\perp, \emptyset)$. The correct chain applies M to the previous connecting map; the two first differ at stage $1 \rightarrow 2$, where both values lie among §7.4’s five elements of I^{++}, so Figure 4 does not discriminate them and the 1, 2, 5, 20 count is unaffected. r0037 reduced the second sentence to one proposition, <code>LemThirty.Thm29Normal</code> (a normal embedding $K(E) \rightarrow A^\infty$), with <code>exists_embeddingProjectionPair_of_thm29Normal</code> deriving the whole sentence from it in ~60 lines — order-reflection buys $p \circ g = \text{id}$, normality buys directedness of $f^{-1}(J)$. Two further corrections: the missing step is not extending <code>Stg n</code> to <code>Stg (n+1)</code>, which <code>exists_stage_ge_of_finite</code> proves (stages are already cofinal among finite subsets of A^∞) — it is getting $K(E)$ into A^∞ at all; and r0036’s <code>Colimit.Thm29Second</code> is stronger than the printed sentence, since <code>countable_compacts_of_reflects</code> forces $K(E)$ countable while an uncountable flat cpo is bifinite, so dropping the paper’s word “domain” makes it false rather than open. <code>Thm29SecondAtDomains</code> restores the hypothesis. That last refutation is prose in the docstring, not kernel-checked — the countability lemma is checked, the step from it is not, the same gap <code>Thm 26</code>’s arity-0 argument has

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
7	Lem	30	Ten operators — Lemma 28’s nine plus $()^{\natural}$, the convex powerdomain, which is the whole point of §7.4 — p-representable over the universal bifinite V. An earlier draft of this row said nine, copying Lemma 28’s list	✗ prove — 0 of 10, but statable for the first time: V was built in r0036 (Colimit.V, with $V \approx_o V^+$), and the blocker this row recorded is gone. Only the $\rightarrow$ conjunct is currently type-correct, carried as Colimit.Lem30Arrow; the other nine do not yet exist as $Cpo \rightarrow Cpo$ operators in the Fp setting. The notion itself exists: PRepresentable.IsPRepresentable/IsPRepresentable₂ over $\mathfrak{t}(Fp\ U)$ (r0034), with eq_id_of_mem_Fp_of_mem_Fc proving Fp and Fc meet only at ScottHom.id — so the distinction from Lemma 28’s IsRepresentable is kernel-checked, not merely documented. Also landed: isProjection_repOf, the projection half of the paper’s conjugation recipe. No longer blocked behind the unobtainable [Gun87] — V was built directly. r0037 states it in full: LemThirty.Lemma30 is one ten-fold conjunction with lemma30_of taking ten named hypotheses, so the count is kernel-checked, and lemma30_iff_lemma28_and_plotkin turns “Lemma 28’s nine plus $()^{\natural}$” from prose into a theorem. plotkinOp is defined; its [Domain D] is spent in exactly one place, Plotkin.FinCompacts.instCountable. The cost is measured and it is not ten fresh proofs: PRep’s schemes are generic in the carrier — nothing but the Fp interface plus the retraction pair — so instantiating at V is one obtain and one exact each, and every pair is Theorem 29’s second sentence at $E := F(V)$ with Lemma 17 supplying bifiniteness. Lemma 17’s ten conjuncts are exactly Lemma 30’s ten operators

Tally (matched): ✓ reuse Mathlib $\approx$ **12** (poset, directed, cpo, $\perp$, monotone, continuous, fixed-point Thm 1 & operator, Schröder–Bernstein, algebraic lattice, product, λ -notation) · $\sim$ partial $\approx$ **4** (compact element, sum/lift, ideal completion Thm 11, Thm 22) · $\times$ define / prove $\approx$ **44**, of which 4 are done ($\ll$ r0003; algebraic, domain, bounded-complete r0004) $\rightarrow$ **40 remaining** — the bifinite / powerdomain / D^∞ development and all 28 numbered results.

What is done, and what is next, is listed under **Progress** at the top of this file.